# INDIA'S CHIP MOMENT

## An entrepreneur's take by InCore Semi's GS Madhusudan

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For decades, India's relationship with semiconductors has been largely cerebral – we designed, we coded, we architected for the world. But we didn't own the stack. Not fully, and never end to end. GS Madhusudan wants to change that.

As CEO and co-founder of InCore Semiconductors, Madhusudan sits at an inflection point in India's semiconductor story – a moment where ambition, policy, and technical maturity are starting to converge. If India's chip dreams are to move from PowerPoint to production, companies like his will have to carry a disproportionate share of the weight.

InCore wasn't conjured out of thin air. It emerged from the RISE lab at IIT Madras – the same ecosystem that built Shakti, India's first indigenous microprocessor.

"The team came out of the Shakti effort where we had prototyped three chips," Madhusudan tells me. "So we had confidence even before we started InCore." Confidence, in semiconductor terms, is silicon that ultimately works without any fuss.

By the time InCore was formally established in 2018, the founders had already walked the hard road from RTL to tape-out, earning credibility in nanometres and yield percentages. For customers, he says, "our InSoC test SoC would have been a major confidence booster." It wasn't just brilliant engineering, but repeatable engineering that worked.

### What most people don't understand

Ask Madhusudan what India gets wrong about its own semiconductor aspirations, and he doesn't hesitate. "The complexity of the eco-system, the supply chain issues, lack of advanced material sciences in India, manufacturing precision, cost of design required are all not understood very well."

Design talent exists. Everything else, he suggests, must be built. "The challenge is in creating an eco-system from scratch. Apart from design talent, nothing else exists in any significant form."

His prescription is pragmatic: "We need to split the challenge into multiple streams – IP, design, fabless, CAD SW, OSAT, fabs – and do independent roadmaps for each." It's not glamorous. But neither is nation-building, suggests Madhusudan.

On the government's role, Madhusudan is measured but still pragmatic. "Any government can do only so much. The PLI and DLI are the right way to do things." But he's clear about where emphasis should shift. "Going forward less focus on fabs and more on IP, fabless and OSAT. Also focus on getting 4-5 local designs to market ASAP."